SKIDS Edge-Stack

v1 Release

Closed-loop clinical decision support for pediatric screening

Released · 2026-04-17

Audience: clinical · ops · tech · business

Executive summary

From capture to decision-support

Five new capabilities turn SKIDS Screen from a data-capture platform into a closed-loop clinical decision-support system.

Doctors gain context, not clicks

Every observation surfaces top-5 relevant clinical evidence snippets + top-5 similar past cases in one expand, in under 500 ms.

Nothing is lost in transit

Durable workflow records 5 steps per observation; queue DLQs catch every poison message; every AI decision is auditable.

Ops + research unblocked

All 5 canonical analytics tiles (Q1–Q5) live; nightly de-identified Parquet layer ready for external researchers.

What shipped — at a glance

#	Feature	Who benefits	Status
1	Durable screening workflow — 5-step trace per observation	Nurses · Ops	**Live**
2	Sandbox second opinion — budget-capped ONNX re-analysis	Doctors · QA	**DB + UI live; container deferred**
3	Evidence RAG + similar cases — 147 chunks, <500 ms P95	Doctors	**Live**
4	Population health Q1–Q5 + de-identified Parquet	Ops · Leadership · Researchers	**Live**
5	On-device Liquid AI — zero cloud egress, OPFS cached	Nurses · Doctors	**Infra live; weights pending**

Live = ready for clinical use today. Amber = one known unblocker before promotion.

1 · Durable screening workflow

ScreeningObservationWorkflow · Cloudflare Workflows + Queues

The 5 recorded steps

1. **persist**
   INSERT OR REPLACE into Turso
2. **quality-gate**
   confidence · risk · routing
3. **embed**
   bge-small-en-v1.5 via Workers AI
4. **enqueue 2nd opinion**
   cond-based fan-out to queue
5. **await-review → notify**
   72h timeout · doctor-review event

Impact

Before: any transient error mid-flow silently lost the observation's downstream work.

After: every step records to `workflow_events`. Ops can reconstruct any screening's full trajectory.

Rollback: `FEATURE_USE_WORKFLOW=0` → redeploy. Inline path resumes; in-flight workflows still drain.

2 · Sandbox second opinion

Doctor-initiated ONNX re-analysis · budget-capped per session

Triggers

- AI confidence < 0.75
- Moderate-risk finding with confidence < 0.9
- Any vision / ear / skin / dental observation
- Doctor manual click in the inbox

What's deferred

Only the ONNX Docker image. DB schema, queue infra, UI button, session budget — all live. One container deploy enables inference end-to-end.

5
Max second-opinions per session (hard cap)

≥0.80
Target agreement score vs. doctor

60s
P95 budget for queue → ONNX → DB

paired
accuracy_metrics row on every opinion

3 · Evidence RAG + similar cases

GET /api/reviews/:id/context · evidence + similar cases in one fan-out

147 chunks

Cloudflare Vectorize index
384-dim cosine

<500 ms P95

Unified /context endpoint
evidence + similar in parallel

top-5 each

Evidence snippets + similar past cases
per observation

📖 What's indexed today

- 52 4D condition descriptions
- 20 M-CHAT items with domain + criticality
- Per-module parent-education intros
- Condition-level parent guidance

🔒 Privacy posture

- Only curated, pre-approved educational content
- 280-char previews in metadata, no full text
- No PHI, no patient-record free text
- Index versioned in `evidence_index_version`

4 · Population health analytics

Five canonical tiles live on /population-health · Turso hot path + nightly Parquet

Q1 AI ↔ clinician agreement

Catch miscalibration per module × age band early

Q2 AI spend breakdown

Detect cost regressions from a model swap within 7 days

Q3 Red-flag prevalence

Flag unusual campaign × age × gender clusters for follow-up

Q4 Screener throughput

Identify training opportunities or tool friction per nurse

Q5 Time-to-review SLA

P50 / P95 / P99 minutes by decision — SLA monitoring

Analysts can also query the nightly de-identified Parquet directly — no PHI, safe for research.

5 · On-device Liquid AI — LFM2.5-VL-450M

Zero cloud egress · OPFS cached · per-shard SHA-256 verified

Live in production

- Same-origin shard proxy `/api/models/:id/:version/:shard`
- OPFS cache with per-shard SHA-256 verification
- WebLLM + WebGPU runtime handoff
- HITL outcome audit (suggested/accepted/rejected/edited)
- Pinned manifest — no silent model upgrades

Pending dev-team

- LFM2.5-VL-450M weight shards not yet uploaded
- `MODEL_MANIFEST` still `PENDING-PIN` placeholder
- Mobile track (React Native runtime) not started
- Est. 2–4 engineering days to complete
- `docs/HANDOVER-LIQUID-AI.md` is self-contained plan

Safety & governance — shipped alongside

Audit everywhere

audit_log captures every write, AI decision, model-shard fetch, queue message, and failure.

DLQ on every queue

3 retries → dead letter → audit row + optional Langfuse trace. No poison message is silent.

Budget caps

session_ai_budget (per-session) + ai_usage (rollup). Q2 tile surfaces this on the dashboard.

Rollback on every flag

Every new capability is flag-gated in wrangler.toml. Flip to "0" + redeploy = instant revert.

PHI residency

Media stays in R2 APAC. Vectorize holds only curated text. Parent PDFs are HMAC-signed URLs, 30-day TTL.

Every PR verified

Typecheck + tests + /api/health smoke before merge. No deploy without a clean pipeline.

Metrics to watch post-launch

Metric	Source	Target
Observation P95 write latency	AI Gateway · Langfuse	**< 200 ms**
Workflow success rate	`workflow_events` table	**> 99%**
Second-opinion ↔ doctor agreement	`accuracy_metrics.agreement_score`	**> 0.80 rolling**
Evidence search P95	`/api/evidence/search` Langfuse span	**< 200 ms**
/context unified endpoint P95	`/api/reviews/:id/context`	**< 500 ms**
DLQ depth per queue	`wrangler queues consumer get`	**always 0 (alarm > 5)**
Publishable Parquet freshness	R2 `publishable/dt=…`	**one partition per day**

What's next — prioritised

P0	**Liquid AI weights upload** Dev team (handover doc ready)	*2–4 eng days*
P1	**Sandbox AI container image** Ops + platform	*After container beta stable*
P1	**Doctor-inbox 2nd-opinion badge in row header** Frontend	*1 sprint*
P2	**Parent SMS / WhatsApp delivery adapter** Integrations	*Q2*
P3	**scripts/duckdb-repl.sh for researchers** Ops	*Nice-to-have*

Explicitly deferred: DuckDB in-Worker (Phase 08 / MotherDuck). Architectural decision, not a bug.

How this was built

One engineering day — surfaced, fixed, documented end-to-end.

13	7	6	100%
PRs merged today	feature PRs (#22–29)	fix PRs (#30–36)	auto-squash + verified

- Typecheck + tests before every merge
- `/api/health` + canonical-query smoke after every deploy
- Feature flags on every capability — instant rollback path
- Full blueprint + deferred-items register in `docs/BLUEPRINT.md`

Shipped · Auditable · Rollbackable